UniCompass — A Real-Time Opportunity Aggregator & Discovery Engine

Team: 18

1. Description of the Use Case

Users / Stakeholders:

- **B.Tech Undergraduates:** Engineering students looking for internships, freelancing gigs, online courses, and hackathons.
- **Postgraduate Aspirants / Researchers:** Undergraduate students seeking **M.Tech / MS / PhD** programs and active research lab opportunities.

Problem : Career and academic opportunities are **highly fragmented** across thousands of websites and portals. Many postings (hackathons, internships, research openings) expire quickly, making it difficult for students to discover them in time. Maintaining a static database of opportunities is impractical due to the **volume and constantly changing nature** of postings.

Significance : Missing an opportunity due to lack of awareness can impact a student's academic and career growth. **UniCompass solves this by shifting from passive searching to active discovery**, automatically tracking opportunities and notifying students of relevant openings as soon as they appear.

2. Key Functionalities

Core Features

Feature	Description
RSS & Aggregator Engine	Collects opportunities from RSS feeds and job APIs.
Real-Time Alerts	Uses semantic matching to notify users of relevant opportunities.
Profile Matching	Ranks opportunities using user skills and interests.
Discovery Feed	Unified feed to browse, filter, save, and bookmark opportunities.
Resume Profile Builder	Extracts skills from uploaded resumes to create user profiles.

User Interaction: Users access UniCompass through a **responsive web application** with an onboarding form, a personalized opportunity feed, and **real-time notifications** for highly relevant openings.

Technical Stack

- **Frontend:** React, Next.js, TailwindCSS, Socket.io-client
- **Backend:** Python, FastAPI (REST + WebSockets)
- **Data Sources:** RSS feeds (feedparser), Aggregator APIs (Adzuna, Jooble, Semantic Scholar)
- **Databases:** PostgreSQL, Redis, pgvector
- **AI/ML:** Gemini API (resume parsing), Sentence-Transformers (semantic matching)

Design Patterns

Pattern	Purpose
Facade & Adapter	Unifies RSS parsers and APIs into a standard `OpportunityCard` format.

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Observer (Pub-Sub)	Triggers user notifications when new matching opportunities appear.
Strategy	Enables dynamic feed sorting (latest or highest relevance).
Repository	Separates database access from business logic.

Expected Prototype Timeline

Prototype Features:

- RSS Ingestion Engine & Discovery Feed
- Resume Upload & AI Profile Builder

Phase	Duration	Key Activities
Setup & Research	Week 1	Identify RSS sources, design PostgreSQL schema, initialize FastAPI and Next.js.
Data Ingestion & Feed	Week 2	Build RSS worker with feedparser, store data in PostgreSQL, implement discovery feed API & UI.
AI Profile Builder	Week 3	Extract resume text with PyMuPDF and generate structured profiles using Gemini API.
Integration & Testing	Week 4	Connect resume profiles with the feed, test the pipeline, refine UI, and deploy.

Domain: Education / EdTech Discovery

UniCompass operates in the **education opportunity discovery domain**, aggregating internships, hackathons, and research openings through **RSS feeds and APIs**.It provides a **dynamic, personalized feed** where opportunities automatically match students' skills and interests, helping them discover **time-sensitive academic and career opportunities** more efficiently.

Architecture Diagram

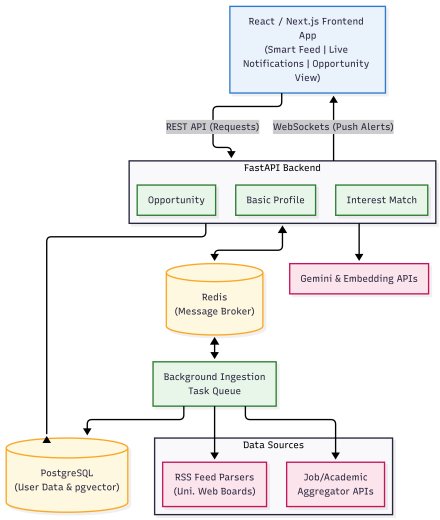

Figure 1: Architecture Diagram